Laboratory 0b11

Expected submission ON PORTALE DELLA DIDATTICA: SURNAME_NAME_MATRICOLA_lab_03.zip

Including:
- program_1
- program_1_r
- program_1_u
- lab_03.pdf (complete this file and export it as PDF)

Submission deadline: 27/10/2026

1) Create a new program (**program_1**) as follows:

Let *v[0..N-1]* be the $N = 4$ single-precision floating-point values provided below. Write the code that normalizes *v* and stores the result in a new vector *bn[0..N-1]*:

$$\mu = \frac{1}{N}\sum_{i=0}^{N-1} v[i] \qquad \text{(mean)}$$

$$\sigma^2 = \frac{1}{N}\sum_{i=0}^{N-1} (v[i] - \mu)^2 \qquad \text{(variance)}$$

$$bn[i] = \gamma \cdot \frac{v[i] - \mu}{\sqrt{\sigma^2 + \varepsilon}} + \beta \qquad \text{(normalization)}$$

Requirements:

- Implement the three steps as three separate loops (mean, variance, normalization) using RISCV floating point instructions, refer to the ISA slides/documentation.
- Compute $\sqrt{(\sigma^2 + \varepsilon)}$ only once, outside the normalization loop, and store it in result (variable in data section).
- Copy the following lines into the `.data` section of `program_1`. They provide the values v, the constants gamma = 1.0, beta = 0.0 and eps = 0.001, and the space for bn and result:

```
# ---------- Batch of N = 4 fixed values ----------
N:      .word 4
v:      .float 0.62, -0.35, 1.18, 0.07
# ---------- Batch normalization ----------
gamma:  .float 1.0
beta:   .float 0.0
eps:    .float 0.001
bn:     .space 16       # N normalized floats
result: .space 4        # sqrt(variance + eps)
```

2) Use the following CPU configuration, run the program (**program_1**), observe the resulting pipeline and answer the following:



a. Manually detect the different hazards that cause pipeline stalls. **Report at least 3 hazards** in the code (or the pipeline) and fill in the following table (first line of the table is filled with a fake example, delete it):

RISC-V Instruction(s)	Type of Hazard	Pipeline Stage	Cause of Stall
i.e. ADD x5, x1, x2 SLLI x6, x5, 1	i.e. Data - RAW	i.e. EX	i.e. x5 not ready for shift until ADD complete
…	…	…	…
fmul.s f10, f2, f10 fadd.s f10, f10, f3	RAW		Here fadd needs f10 and is produced by fmul
fadd.s f7, f7, f5 fsqrt.s f7 , f7	WAW		Fsqrt writs a value that fadd writes also so output dependency
fmul.s f2, f2, f2 fadd.s f5, f5, f2	RAW		Same with the regiser f2

b. Optimize the program (**program_1**) by re-scheduling instructions to eliminate as many hazards as possible. Manually calculate the number of clock cycles for the new program (**program_1_r**) to execute and compare the results with those obtained by the simulator.

c. Unroll the loops of **program_1** with an unrolling factor of 2 (for every loop). Manually calculate the number of clock cycles to execute the new program (**program_1_u**) and compare the results obtained with those obtained by the simulator.

Complete the following table with the results obtained:

Program	program_1	program_1_r	program_1_u
Clock cycles by hand	422	396	377
Clock cycles by simulation	422	396	358

3) Collect the Cycles Per Instruction (CPI) from the simulator for different programs

	program_1	program_1_r	program_1_u
CPI by hand	2.24	2.89	3.20

CPI by simulation	2.24	2.89	3.20

Compare the results obtained and provide some explanation if the results are different.

Explanation (if any):

"To stall, or not to stall: that is the hazard."
— William Shakespeare, Hamlet (RISC-V edition)